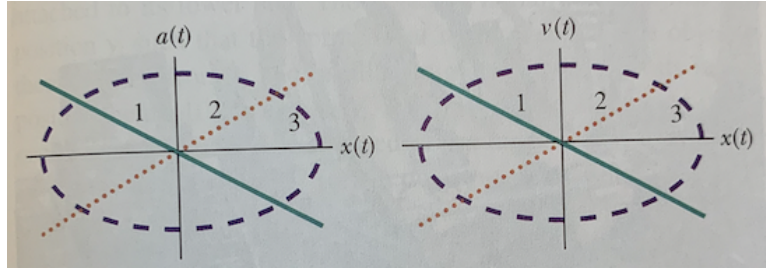


1 Some Quick Conceptual Questions [15 points]

(a) Simple Harmonic Motion [5 points]

Consider a particle executing one-dimensional simple harmonic motion (SHM) without dissipation. The force on the particle is according to Hooke's law,



$$F(x) = -kx$$

which combined with Newton's law

$$ma = m\ddot{x} = -kx$$

$$a = \ddot{x} = -\frac{k}{m}x$$

(i) **Acceleration versus position:** From the equation of motion,

$$a = -\omega^2 x.$$

Thus, acceleration is linearly proportional to position and opposite in sign. The a - x plot is therefore a straight line through the origin with a negative slope. **Hence, curve 1 corresponds to $a(t)$ versus $x(t)$.**

(ii) **Velocity versus position:** As we know solution of the force second order differential equation can be written as

$$x(t) = A \cos(\omega t + \phi),$$

where A is the amplitude, ω the angular frequency and ϕ is the phase at time t . We know the velocity $\dot{x}$ is going to be $\pi/2$ out of phase with a sinusoidal displacement. Just with this, we can argue that **curve 3 corresponds to $v(t)$ versus $x(t)$.**

For a more detailed answer, we see that the velocity and acceleration are

$$v(t) = \dot{x}(t) = -A\omega \sin(\omega t + \phi),$$

$$a(t) = \ddot{x}(t) = -\omega^2 x(t).$$

Eliminating time between $x(t)$ and $v(t)$,

$$x = A \cos(\omega t + \phi), \quad v = -A\omega \sin(\omega t + \phi).$$

Squaring and adding both,

$$\frac{x^2}{A^2} + \frac{v^2}{A^2\omega^2} = 1.$$

This is the equation of an ellipse in the $x-v$ plane.

Rubric: 2.5 points per sub-part. Full points if solution includes a justification directly linking a and v with position based on basic SHM arguments. 1 point per sub-part if correct answer but misses/incomplete justification. 0.5 point each sub-part for incorrect answers but for an attempt.

(b) Roller Coaster Tracks and Small Oscillations [5 points]

The two tracks are given by (with appropriate units):

$$\text{Track 1: } x^2 + (y - 1)^2 = 1,$$

$$\text{Track 2: } x^2 + 4(y - 1)^2 = 4.$$

as we can see, both have their lowest point at $(0, 0)$.

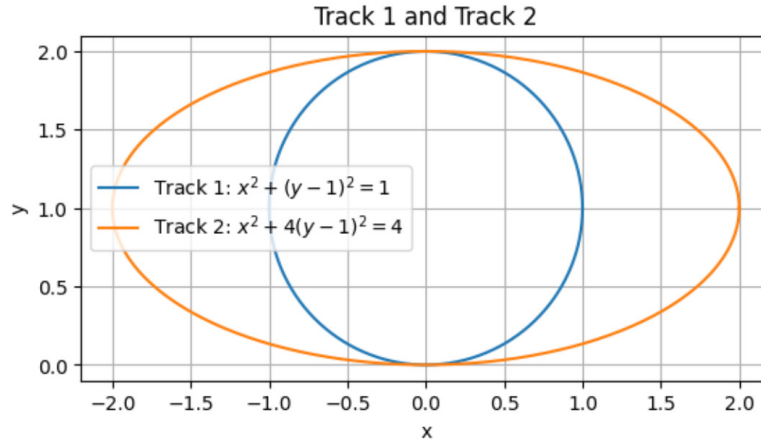


Figure 1: Plot of Track1 and Track2

Consider a one-dimensional system of mass m moving in a potential $U(x)$. The equation of motion is given by Newton's law,

$$m\ddot{x} = -\frac{dU}{dx}. \quad (1)$$

since $x = 0$ is point of stable equilibrium. Then,

$$\left. \frac{dU}{dx} \right|_{x=0} = 0, \quad \left. \frac{d^2U}{dx^2} \right|_{x=0} > 0. \quad (2)$$

Expand the potential $U(x)$ in a Taylor series about $x = 0$:

$$U(x) = U(0) + \left. \frac{dU}{dx} \right|_{x=0} x + \frac{1}{2} \left. \frac{d^2U}{dx^2} \right|_{x=0} x^2 + \dots$$

neglecting higher order term and using the equilibrium condition $\frac{dU}{dx}\big|_{x_0} = 0$, this reduces to

$$U(x) \approx U(0) + \frac{1}{2} \frac{d^2U}{dx^2}\bigg|_{x=0} x^2.$$

considering potential at $x = 0$ as zero i.e $U(0) = 0$; (it's our choice, where we want our zero potential)

The force is therefore

$$F = -\frac{dU}{dx} \approx -\frac{d^2U}{dx^2}\bigg|_0 x = -kx = m\omega^2 x. \quad (3)$$

which implies that:

$$\boxed{\omega^2 \propto \frac{d^2U}{dx^2}\bigg|_{x=0}}.$$

Thus, **for small oscillations about a stable equilibrium, the oscillation frequency is determined by the curvature of the potential energy at the equilibrium point.**

A larger curvature corresponds to a stronger restoring force and hence a higher oscillation frequency. From the above Fig. 1, clearly **curve 1** is steeper (more curvature) so it corresponds to higher frequency. **Therefore, Track 1 has the higher oscillation frequency.**

Also, we can compute the value of the frequency using the second derivatives of the given curves,

The two tracks are given by:

$$\begin{aligned} \text{Track 1: } x^2 + (y - 1)^2 &= 1, \\ \text{Track 2: } x^2 + 4(y - 1)^2 &= 4. \end{aligned}$$

Both have their lowest point at $(0, 0)$.

For small oscillations, the motion is determined by the curvature of the potential energy near the equilibrium point.

Track 1

From the equation of the track,

$$(y - 1)^2 = 1 - x^2.$$

Taking the lower branch and expanding for small x ,

$$y = 1 - \sqrt{1 - x^2} \approx \frac{x^2}{2}.$$

The gravitational potential energy is (dropping multiplicative terms with dimensions and only looking at the scaling)

$$U = mgy \approx \frac{mg}{2} x^2 \propto \frac{1}{2} k_1 x^2.$$

Track 2

Similarly,

$$(y - 1)^2 = 1 - \frac{x^2}{4}.$$

For small x ,

$$y = 1 - \sqrt{1 - \frac{x^2}{4}} \approx \frac{x^2}{8}.$$

Thus,

$$U = mgy \approx \frac{mg}{8}x^2 = \frac{1}{2}k_2x^2.$$

hence $\boxed{k_2 = \frac{k_1}{4}}$

For small oscillations,

$$\omega = \sqrt{\frac{k}{m}},$$

which gives us $\omega_1 > \omega_2$ by a factor of 2. Therefore, **Track 1 has the higher oscillation frequency.**

Rubric: Full 5 points for correct answer with complete justification/brief derivation connecting frequency of oscillations with curvature of potential at equilibrium. 2.5 points if correct answer but missing/incomplete justification (including just saying track 1 is steeper by inspection). Frequency calculation is not needed explicitly. 1 point if incorrect or if something is attempted.

(c.) Swing Being Periodically Pushed [5 points]

A playground swing of length $l = 2.5$ m carrying a child of mass $m = 30$ kg can be modeled as a simple pendulum. For small oscillations, its equation of motion (including damping and a periodic driving force) is

$$\ddot{\theta} + \gamma\dot{\theta} + \omega_0^2\theta = \frac{F_0}{ml}\cos(\omega t), \quad (4)$$

where

$$\omega_0 = \sqrt{\frac{g}{l}} \quad (5)$$

is the natural angular frequency of the pendulum and γ is the damping coefficient.

As derived in class, for a driven, damped harmonic oscillator, the steady-state amplitude is maximized when the driving angular frequency equals the resonant frequency

$$\boxed{\omega_m = \sqrt{\omega_0^2 - \frac{\gamma^2}{2}}} \quad (6)$$

This result follows from maximizing the amplitude response

$$A(\omega) \propto \frac{1}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}}$$

with respect to ω .

Since air resistance and other damping effects are negligible,

$$\gamma \ll \omega_0,$$

and therefore

$$\omega_m \approx \omega_0. \quad (7)$$

Thus, maximum energy transfer occurs when the adult pushes the swing at its natural frequency (**resonance condition**).

Using $g \approx 10 \text{ m/s}^2$,

$$\omega_0 \approx \omega_m = \sqrt{\frac{g}{l}} = \sqrt{\frac{10}{2.5}} = 2 \text{ rad/s}. \quad (8)$$

The corresponding frequency is

$$f = \frac{\omega_0}{2\pi} = \frac{2}{2\pi} = \frac{1}{\pi} \text{ Hz}.$$

Hence, the number of pushes per minute for maximum effectiveness is (using $\pi \approx 3$)

$$N = 60f = \frac{60}{3} = 20 \text{ pushes per minute}. \quad (9)$$

So for maximum effectiveness, the adult should push the swing approximately

$$20 \text{ times per minute}$$

in phase with the swing's motion.

Rubric: Full 5 points for correct answer with complete justification of identifying resonance condition, arguing for small damping, and getting correct result. 2.5 points if correct answer but missing/incomplete justification, including if small γ not discussed. 1 point will be taken away for numerical error (and even if ω_0 is taken to be $\sqrt{l/g}$ instead of $\sqrt{g/l}$) while rest is consistent, if any, over and above the given rubric. 1 point overall if incorrect answer but something is attempted.

2 NASA Wants Your Help! [10 points]

a Mass measurement using equation of motion (BMMD) [2 points]

Consider an astronaut of mass M sitting on a spring-mounted chair of effective mass m . The total oscillating mass is therefore

$$m_{\text{tot}} = M + m$$

When displaced slightly from equilibrium, the restoring force due to the spring is given by Hooke's law:

$$F = -kx$$

Using Newton's second law,

$$m_{\text{tot}}\ddot{x} = -kx,$$

or

$$(M + m)\ddot{x} + kx = 0$$

This is the standard equation of simple harmonic motion:

$$\ddot{x} + \frac{k}{M + m}x = 0$$

$$\ddot{x} + \omega^2 x = 0$$

where, the angular frequency, ω is given by

$$\omega = \sqrt{\frac{k}{M + m}}$$

The time period of oscillation is therefore

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{M + m}{k}}$$

Squaring both sides:

$$T^2 = 4\pi^2 \frac{M + m}{k}$$

Rearranging to find the astronaut's mass:

$$M + m = \frac{kT^2}{4\pi^2},$$

$$\boxed{M = \frac{kT^2}{4\pi^2} - m}$$

Hence, by measuring the oscillation period T and knowing the spring constant k and effective chair mass m , the astronaut's mass can be determined.

Rubric: Full 2 points for correct answer with justification including invoking SHM, writing basic equation of mass on spring and connecting the time period with total mass, and proper algebraic rearrangement. 1 point for correct answer but missing steps/lack of justification of final expression. 0.5 point if incorrect but something is attempted.

b Mass of empty chair [1 point]

As we know,

$$M + m = \frac{kT^2}{4\pi^2},$$

For an empty chair, $M = 0$, so eq. becomes

$$m = \frac{kT^2}{4\pi^2},$$

Given:

$$k = 40\pi^2 \text{ N/m}, \quad T = 1 \text{ s}$$

Substituting the values:

$$m = \frac{40\pi^2 \times (1)^2}{4\pi^2}$$

Canceling π^2 :

$$m = \frac{40}{4} = 10 \text{ kg}$$

$$\boxed{m = 10 \text{ kg}}$$

Thus, the effective mass of the empty chair is 10 kg.

Rubric: Full 1 point if proper variables are plugged to obtain answer with appropriate units. 0.5 point if units are missing or lack of astronaut M is not taken into account or numerical error. 0 points in all other cases.

c **Mass of astronaut Alan Bean** [1 point]

Given:

$$k = 40\pi^2 \text{ N/m}, \quad m = 10 \text{ kg}, \quad T = 3 \text{ s}$$

As,

$$M + m = \frac{kT^2}{4\pi^2}$$

Substituting the values:

$$M + 10 = \frac{40\pi^2 \times 9}{4\pi^2}$$

Canceling π^2 :

$$M + 10 = \frac{360}{4} = 90$$

Thus,

$$M = 90 - 10 = 80 \text{ kg}$$

$$\boxed{M = 80 \text{ kg}}$$

Hence, the mass of astronaut Alan Bean is approximately 80 kg.

Rubric: Full 1 point if proper variables are plugged to obtain answer with appropriate units. 0.5 point if units are missing or numerical error. 0 points in all other cases.

d **Minimum astronaut mass for sustained oscillations** [3 points]

The equation of motion of the damped BMMD system is

$$(M + m)\ddot{x} + b\dot{x} + kx = 0.$$

Dividing throughout by $(M + m)$:

$$\ddot{x} + \frac{b}{M + m}\dot{x} + \frac{k}{M + m}x = 0.$$

Define

$$\gamma = \frac{b}{M + m}, \quad \omega_0^2 = \frac{k}{M + m}$$

Thus the equation becomes

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

Assuming solution $x = e^{rt}$:

$$r^2 + \gamma r + \omega_0^2 = 0$$

The roots are

$$r = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2}$$

Condition for underdamped oscillations

For sustained oscillations,

$$\gamma^2 < 4\omega_0^2$$

Substituting back:

$$\left(\frac{b}{M + m}\right)^2 < 4\frac{k}{M + m}$$

Multiplying by $(M + m)^2$:

$$b^2 < 4k(M + m)$$

Minimum astronaut mass

$$M + m > \frac{b^2}{4k},$$

$$M > \frac{b^2}{4k} - m$$

Hence, the minimum astronaut mass measurable while keeping the system underdamped is

$$M_{\min} = \frac{b^2}{4k} - m.$$

Rubric: Full 3 points if condition of underdamping is used while properly defining γ and ω_0 and justifiably shown the algebraic manipulations to get to final inequality. Not necessary to prove $\gamma < 2\omega_0$ from first-principle. 1.5 points if steps are missing or justification is incomplete/inconsistent. 0.5 if incorrect but something is attempted.

e Underdamped frequency and time period [3 points]

As,

$$r = -\frac{\gamma}{2} \pm i\sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$r = -\frac{\gamma}{2} \pm i\omega_d$$

Thus, the underdamped angular frequency, ω_d is

$$\omega_d = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

Substituting

$$k = 40\pi^2, \quad b = 60\pi, \quad M + m = 90 \text{ kg},$$

we get

$$\omega_0^2 = \frac{40\pi^2}{90} = \frac{4\pi^2}{9},$$

$$\frac{\gamma^2}{4} = \left(\frac{b}{2(M+m)}\right)^2 = \left(\frac{60\pi}{180}\right)^2 = \frac{\pi^2}{9}$$

Hence,

$$\omega_d = \sqrt{\frac{4\pi^2}{9} - \frac{\pi^2}{9}} = \frac{\pi}{\sqrt{3}} \text{ rad/s}$$

The damped time period is

$$T_d = \frac{2\pi}{\omega_d} = 2\sqrt{3} \text{ s}$$

Using $\sqrt{3} \approx 1.7$, we get

$$T_d \approx 3.4 \text{ s}$$

Rubric: Full 3 points if a change in time period is justified through modification of oscillation frequency to ω_u , along with full steps in calculation. Not necessary to derive ω_u from first principle. 1.5 points if correct answer but the use of ω_u is not justified/cited. 1 mark taken away for numerical error and/or missing units. 0.5 point for incorrect answers but if something is attempted.